Flexible Hurricane Protection

Parametric Insurance Product Design

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Overview

The parametric risk product for Flexible Hurricane Protection has been developed by London-based parametrics insurance specialists Global Parametrics. They have undertaken to price the risk, and to develop policy options that minimize basis risk while also keeping prices affordable. CREAD has worked with GP to develop policy options that address the following challenges:

- **Minimizing product complexity** – parametric insurance has never been sold at the retail level in the region, so keeping the product design simple and comprehensible is a priority.
- **Keeping premiums affordable** – there is a natural tension of course between minimizing basis risk and keeping premiums affordable. The policies have been designed in such a way that keeps premiums below 15% of the coverage amount, while also endeavoring to ensure that basis risk is kept as low as possible.
- **Minimizing basis risk** – a key priority is to ensure that the risk of damage and triggering events are as closely correlated as possible, to avoid as much as possible the situation where a storm occurs, but policy holders don't receive a payout.

Product Design

The risk product that GP has developed is what is referred to as a "cat in a box", shorthand for "catastrophe in a box". This refers to the fact that the parametric product measures specific weather conditions in defined locations.

"Cat in a box" Product

Cat in a box products effectively demarcate a particularly set of geographical boundaries, defines one or more specific categories of perils to be monitored for, sets the triggering parameters, and defines a coverage period for the policy. The specific parameters are discussed below, but it is worth indicating the positives as well as the negatives associated with this type of product design, along with a discussion on why a single peril was selected.

Selection of Peril

We have chosen to build the product around a single peril, rather than blending perils such as excess rainfall and windspeed. While it is undoubtedly true that the greatest amount of damage from tropical cyclones is as a result of excess rainfall, we determined that adding additional perils would adversely impact product pricing and complexity. Selling to retail customers is not the same as selling to governments and other sophisticated buyers, and a parametric policy built around a series of complex and opaque triggers was not appropriate for the market or for the pilot.

To adjust for this, it is our intention throughout the sales cycle to make buyers aware of the fact that they are buying protection specifically against the *occurrence* of tropical cyclone strength winds – they are not buying protection against potential damages that might result from a tropical cyclone.

Impact on basis risk

Cat in a box products do not perfectly correlate potential damage incurred with the measured climate event. That is – measuring the catastrophe is an imperfect proxy for whether or not an individual actually incurs damage. The product as designed measures only wind speed, which increases basis risk (basis risk can be defined as the chance that an event will occur that doesn't trigger a payout but still causes damage, or that a payout is triggered even when no damage has been incurred).

Adding additional perils would reduce basis risk, but given the impact on product complexity and pricing, we opted not to include additional perils for the pilot.

Impact on complexity

It would be possible to reduce basis risk by adding additional perils. For instance, CCRIF measures wind speed, rainfall, and has an algorithm for anticipating economic loss. But CCRIF is sold to Ministries of Finance, not to rural farmers. Therefore, we took a view that adding additional perils (e.g., rainfall) would increase the complexity of the product, as well as driving up the price. We kept the peril covered by the cat-in-a-box design to one, and we kept the triggering parameters transparent. Increasing complexity by adding perils would also reduce basis risk, but for the reasons mentioned above we kept the peril covered to one.

Impact on price

Adding additional perils would reduce basis risk, as noted above. However, because of the adverse impact on pricing, we determined that we would keep the perils covered to just tropical cyclone.

Triggering Parameters

For FHP, the triggering parameters are storm strength and distance of the eyewall of the storm from the shoreline of Dominica. These are the obvious triggers for a product built around wind speed and have the added advantage of being familiar to consumers.

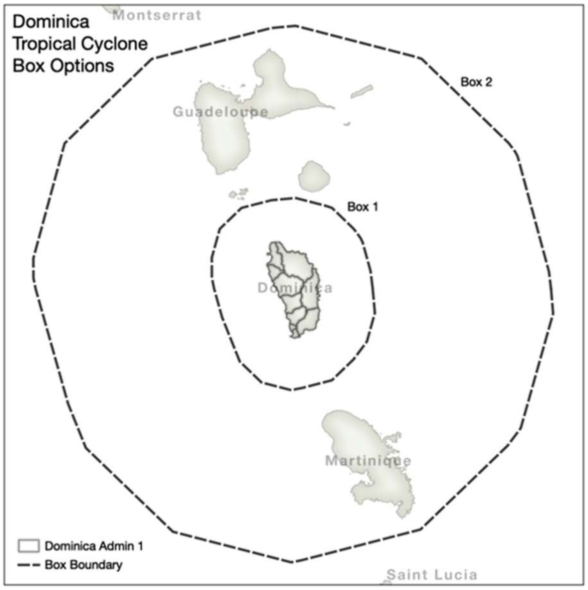
Storm Strength

FHP policies will be built around recognizable storm strength identifiers. These are easy for policy holders to comprehend, transparent in that the data on storms is widely available and allow for a wide range of policy options. Since tropical cyclone strength is a measure of wind speed, the policies will be built around the following parameters.

Tropical Storm	From (and including) 18 m/s and less than 33 m/s
Category 1	From (and including) 33 m/s and less than 43 m/s
Category 2	From (and including) 43 m/s and less than 50 m/s
Category 3	From (and including) 50 m/s and less than 58 m/s
Category 4	From (and including) 58 m/s and less than 70 m/s
Category 5	From (and including) 70 m/s

Distance from Shore

The second parameter that FHP measures as part of the "cat in a box" product is distance of the windward eye wall of the storm from the shoreline of Dominica. The covered region will be divided into two boxes, with the first box 25 kilometers from the shoreline, and the second box 100 kilometers from the shoreline. The diagram below depicts the polygons that include the covered areas. Tropical Storms, and Hurricanes of categories 1 – 3 must pass through box 1. Hurricanes of categories 4 and 5 must pass through box 1 or 2.



The precise definitions of the two polygons are as follows:

	Box 1 25 km		Box 2 100 km	
	lon	lat	lon	lat
1	-61.0007	15.4297	-60.1927	15.427
2	-60.9908	15.3197	-60.1833	15.3169
3	-61.0271	15.19	-60.3301	14.7985
4	-61.0871	15.11	-60.3903	14.7187
5	-61.1856	15.0151	-60.7843	14.34
6	-61.32	14.9804	-61.3202	14.2016
7	-61.37	14.9704	-61.3699	14.1916
8	-61.5044	15.0051	-61.9058	14.33
9	-61.6029	15.1001	-62.2997	14.7088
10	-61.683	15.3	-62.3804	14.9085
11	-61.7193	15.4297	-62.5273	15.427
12	-61.7294	15.4797	-62.5375	15.4769
13	-61.7295	15.5697	-62.538	15.5669
14	-61.6936	15.6996	-62.3962	16.0868
15	-61.595	15.7947	-62.0019	16.4684
16	-61.4601	15.8296	-61.4605	16.6084
17	-61.35	15.8396	-61.35	16.6184
18	-61.2151	15.8048	-60.8085	16.4786
19	-61.1851	15.7948	-60.7784	16.4686
20	-61.0864	15.6996	-60.384	16.0871
21	-61.0565	15.6596	-60.3541	16.047
22	-61.0465	15.6396	-60.3441	16.0269
23	-61.0106	15.5097	-60.2023	15.507
24	-61.0007	15.4297	-60.1927	15.427

Tiered Payouts

In order to match payouts with damage, policies will be structured to provide a tiered payout structure, as indicated for each of the six policy options in the table below. It should be noted that multiple tiered options were provided in order to keep the price of the premiums down, and to allow for a range of policy options. During the product design, this was adjusted to increase the payouts for Category 2 storms. This structure reflects the probability of the underlying events, and as such is not subject to manipulation.

Policy Options

Triggering Parameters	T	1	2	3	4	5
• Tropical Storm • Box 1	10%	X	X	X	X	X
• Category 1 • Box 1	25%	25%	X	X	X	X
• Category 2 • Box 1	35%	35%	35%	X	X	X
• Category 3 • Box 1	50%	50%	50%	50%	X	X
• Category 4 • Boxes 1, 2	50%	50%	50%	50%	50%	X
• Category 5 • Boxes 1, 2	100%	100%	100%	100%	100%	100%

Policy Options

In order to offer policies with a range of options to meet individuals risk preferences, we have defined six policy options, which allow individuals to select the coverage for the events which they feel are most likely to pose a risk to the safety and security of income, assets, or agricultural holdings that the policy holder wishes to protect.

General Principles

Policies are structured so that they provide protection for storms of strength equal to and of greater strength than the selected policy option. Therefore, Policy Option T is the most comprehensive since it provides coverage for storms of tropical storm strength and greater.

Premiums

Premiums have been constructed to reflect the on-line rate provided by the underwriter, as well as to allow for a commission to be charged by the reseller, in this case the local credit unions. The commission for the reseller has been set at 20% of the on-line rate. For example:

On-Line Rate	Reseller Premium (20%)	Total Premium
9%	1.8%	10.8%

Note on Pricing

The product pricing consists of two components – the On-Line Rate, and the Reseller Premium. These are discussed belowl

On-Line Rate

The On-Line rate is the premium supplied to us by Global Parametrics and reflects their proprietary pricing of the risk of the underlying event. It reflects a pricing of the risk of the underlying event.

Reseller Premium

The reseller premium of 20%, which is generally in line with industry standard practice, provides cover for operational costs, sales and promotional costs, as well as customer support and administrative costs. It was intended to ensure that the premium for the pilot reflects as close as possible the actual market price of the product, so that the price wasn't artificially suppressed. It does not reflect recovery costs for the technology platform. Subsequent (post pilot) commercial discussions will determine the split of the 20% reseller premium among various upstream participants, as well as whether the 20% surcharge should be increased; changed to a percent charge on the nominal coverage amount; or reduced.

Policy Descriptions

Following is a detailed description of each of the policy options that are available as part of the pilot launch of FHP. Note that the premiums as presented are inclusive of the on-line rate as well as the reseller premium of 20%.

Policy Option T

Premium: **11.64% of Coverage Amount**

- Policy Option T is the most comprehensive coverage.
- With Policy Option T, if a Tropical Storm passes through Box 1, the policy will pay out 10% of the Coverage Amount.
- There is no payout for Tropical Storms passing through Box 2 under any Policy Option.
- Tropical Storms are only covered under Policy Option T.
- Category 1 Hurricanes that pass through Box 1 trigger a 25% payout.
- Category 2 Hurricanes that pass through Box 1 trigger a 35% payout.
- Category 3 Hurricanes that pass through Box 1 trigger a 50% payout.
- Category 4 Hurricanes that pass through Boxes 1 and 2 trigger a 50% payout.
- Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Option 1

Premium: **11.28% of Coverage Amount**

- Tropical Storms are not covered under Policy Option 1.
- Category 1 Hurricanes that pass through Box 1 trigger a 25% payout.
- Category 2 Hurricanes that pass through Box 1 trigger a 35% payout.
- Category 3 Hurricanes that pass through Box 1 trigger a 50% payout.
- Category 4 Hurricanes that pass through Boxes 1 and 2 trigger a 50% payout.
- Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Option 2

Premium: **9.72% of Coverage Amount**

- Tropical Storms are not covered under Policy Option 2.
- Category 1 Hurricanes are not covered under Policy Option 2.
- Category 2 Hurricanes that pass through Box 1 trigger a 35% payout.
- Category 3 Hurricanes that pass through Box 1 trigger a 50% payout.
- Category 4 Hurricanes that pass through Boxes 1 and 2 trigger a 50% payout.
- Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Option 3

Premium: **8.64% of Coverage Amount**

- Tropical Storms are not covered under Policy Option 3.
- Category 1 Hurricanes are not covered under Policy Option 3.
- Category 2 Hurricanes are not covered under Policy Option 3.

- ⇨ Category 3 Hurricanes that pass through Box 1 trigger a 50% payout.
- ⇨ Category 4 Hurricanes that pass through Boxes 1 and 2 trigger a 50% payout.
- ⇨ Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Option 4

Premium: **7.44% of Coverage Amount**

- ⇨ Tropical Storms are not covered under Policy Option 4.
- ⇨ Category 1 Hurricanes are not covered under Policy Option 4.
- ⇨ Category 2 Hurricanes are not covered under Policy Option 4.
- ⇨ Category 3 Hurricanes are not covered under Policy Option 4.
- ⇨ Category 4 Hurricanes that pass through Boxes 1 and 2 trigger a 50% payout.
- ⇨ Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Option 5

Premium: **3.12% of Coverage Amount**

- ⇨ Policy Option 5 provides only catastrophic coverage for Category 5 Hurricanes.
- ⇨ Tropical Storms are not covered under Policy Option 5.
- ⇨ Category 1 Hurricanes are not covered under Policy Option 5.
- ⇨ Category 2 Hurricanes are not covered under Policy Option 5.
- ⇨ Category 3 Hurricanes are not covered under Policy Option 5.
- ⇨ Category 4 Hurricanes are not covered under Policy Option 5.
- ⇨ Category 5 Hurricanes that pass through Boxes 1 and 2 trigger a 100% payout.

Policy Selection Options

In order to reflect the approach consumers are likely to take in purchasing parametric insurance, and in order to maximize the flexibility that parametric insurance can offer, policies are presented in such a way that consumers can purchase policies in one of two ways: 1) by selecting the amount of coverage that they would like to purchase, based on a maximum payout amount which would be received for a Category 5 storm, or 2) by indicating how much they can afford in annual premiums, and seeing how much coverage they can get under various options.

Making Policy Selections by Coverage Amount

Individuals making policy selections by choosing the coverage amount will have approached risk management by thinking about the specific things that are at risk, whether income, agricultural assets, business assets, or other things that need to be protected from tropical cyclones. The coverage amount that is selected reflects the 100% payout that is received under all policy options for a category 5 storm. For the same amount of coverage, premiums will vary based on the policy option selected – the more events covered, the higher the premiums. The affordability question that individuals must make is on whether they would prefer to have less coverage but for more events, or a larger payout for fewer events.

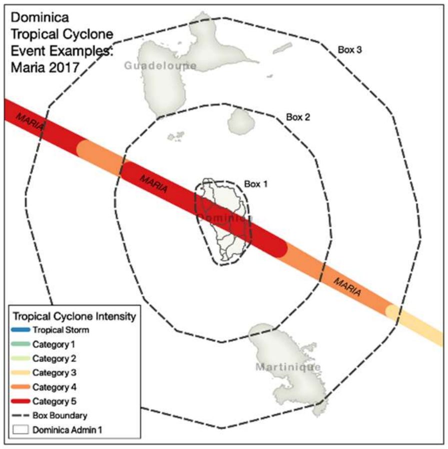
Making Policy Selections by Affordability

By selecting a policy based on the amount that an individual can afford, this gives the consumer the ability to ensure that they have some level of coverage, based on their perceived risk and the amount they can afford to pay. With traditional insurance, consumers don't have the flexibility of adjusting the coverage amount to reflect how much they can afford to pay. This is one of the big benefits of FHP and encourages individuals to think carefully about the risks that they perceive to be most threatening, as well as to the coverage that they can afford. The affordability question that consumers are faced with under this scenario is to simply decide, based on the amount that they can afford, whether they would prefer to have a larger payout for fewer events, or smaller payouts for a wider range of events.

Regulatory Approval

The Financial Services Unit, which is the independent insurance regulator for the Commonwealth of Dominica, provided an exception for the sale of the product for the purposes of the pilot. The exception was granted to the Dominica Cooperative Societies League Ltd, and the exception allowed for the sale of the product only to the two pilot communities, Grand Bay and Salisbury. The regulator was engaged extensively throughout the process, and kept informed on all aspects of product development, market engagement, and technology development. In addition, Global Parametrics was submitted to thorough due diligence.

Appendix I - Historical Storm Data

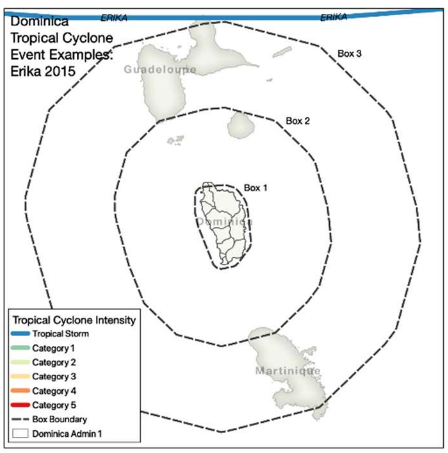


Hurricane Maria

2017

Category 5

Hurricane Maria was a direct hit to Dominica as a category 5 storm, causing damages estimated at EC$2.51 billion and losses of EC$1.03 billion. Losses represented 226% of 2016 GDP. Maria left 31 persons dead and 37 missing.

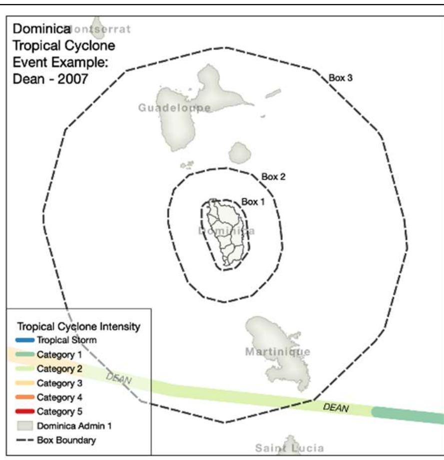


Tropical Storm Erika

2015

Tropical Storm

Erika resulted in heavy rainfall which led to catastrophic flooding and mudslides. Erika was responsible for 30 deaths. 271 homes were destroyed, and 574 persons were left homeless. The entire village of Petite Savanne was evacuated and subsequently abandoned. Damage to infrastructure of approximately EC$500 million.

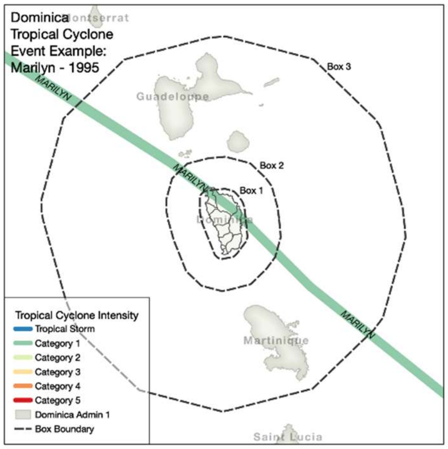
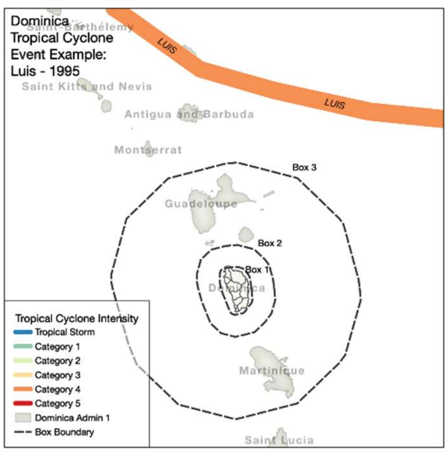
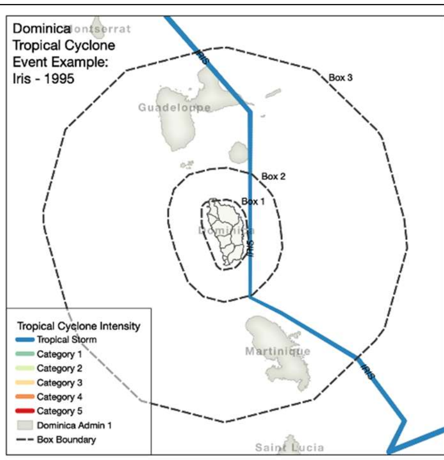


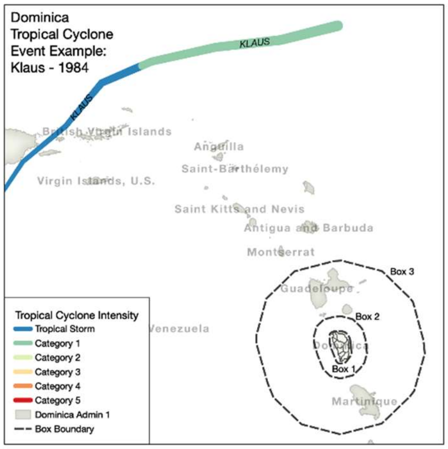
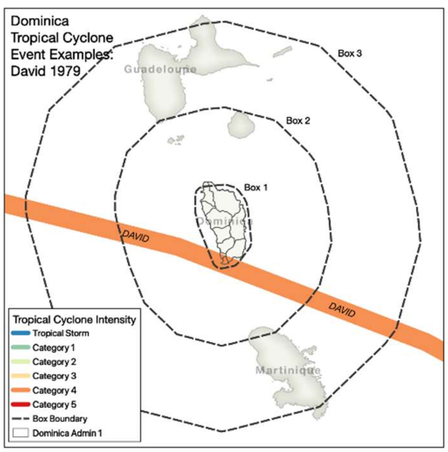
Hurricane Dean

2007

Category 2

Hurricane Dean resulted in 2 deaths and an estimated US$162 million in damages, including 771 buildings that were impacted.

<table>
<tr><td rowspan="4">
</td><td>Hurricane Marilyn</td></tr>
<tr><td>1995</td></tr>
<tr><td>Category 1</td></tr>
<tr><td>Marilyn was the third in a series of storms that hi Dominica in 1995, including Tropical Storm Luis and Hurricane Iris – all three of which struck within 10 days of each other. Marilyn resulted in property damage of approximately EC$184 million.</td></tr>
<tr><td rowspan="4">
</td><td>Tropical Storm Luis</td></tr>
<tr><td>1995</td></tr>
<tr><td>Category 1</td></tr>
<tr><td>Tropical Storm Luis came right on the heels of Tropical Storm Iris. Luis caused serious damage to banana crops and left nearly 1,000 people homeless as a result of rough seas on the western coast. Property damage was an estimated EC$47 million.</td></tr>
<tr><td rowspan="4">
</td><td>Tropical Storm Iris</td></tr>
<tr><td>1995</td></tr>
<tr><td>Category 1</td></tr>
<tr><td>First in a series of three storms that hit Dominica over a span of just 10 days in 1995. Iris caused damage to banana crops, which was then compounded by the subsequent passing of Tropical Storm Luis.</td></tr>
</table>

	**Tropical Storm Klaus**
	1984
	Tropical Storm
	Tropical Storm Klaus impacted 10,000 people in Dominica and resulted in EC$2 million in damage. Two people died as a result of the storm.
	**Hurricane David**
	1979
	Category 4
	Hurricane David grazed the southwestern corner of Dominica with a direct hit. High winds from the category 4 storm severely eroded coastlines and washed-out coastal roads. 56 people died as a result of the storm, and 180 were injured.

Appendix II – FHP Payout Scenarios for Historical Storms

				FHP Payout for Policy Options T – 5 (as % of coverage amount)					
Storm	**Strength**	**Year**	**Passed Through Box**	**T**	**1**	**2**	**3**	**4**	**5**
Maria	5	2017	1	100%	100%	100%	100%	100%	100%
Erika	TS	2015	-	-	-	-	-	-	-
Dean	2	2007	2	-	-	-	-	-	-
Marilyn	1	1995	1	25%	25%	-	-	-	-
Luis	TS	1995	-	-	-	-	-	-	-
Iris	TS	1995	1	10%	-	-	-	-	-
Klaus	TS	1984	-	-	-	-	-	-	-
David	4	1979	1	50%	50%	50%	50%	50%	-

FHP
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